

SPRING BOOT = BACKEND JAVA FRAMEWORK

- WE USE IT TO MAKE REST APIS

API = APPLICATION PROGRAMMING INTERFACE

REST = REPRESENTATIONAL STATE TRANSFER

- DB OBJECTS ARE TREATED AS UNIQUE LOCATIONS ON THE INTERNET

URL - UNIFORM RESOURCE LOCATOR

URI - " " IDENTIFIER

URN - " " NAME

CLIENT/SERVER ARCHITECTURE



FE SERVER → GIVES BROWSER CODE TO RUN

BROWSER → MAKES REQS TO BE

DE → INTERACTS w/ DB

HTTP REQUESTS

↳ HYPERTEXT TRANSFER PROTOCOL

ROUGH SQL EQUIV.

GET [SELECT/READ] - retrieve resource @ location

POST [INSERT/CREATE] - create new record / send data

PUT [UPDATE] - overwrites whole record

PATCH ["] - overwrites only differences

DELETE [DELETE] - REMOVE A RECORD (1 or more)

OPTIONS - GETTING COMMS OPTIONS PRE-REQ

SAFE - DOES NOT MODIFY DB!

UNSAFE - CHANGES DATA AND/OR SCHEMA

IDEMPOTENT - AFTER IDENTICAL REQS,
DB IS IN IDENTICAL STATE

GET - ✓

DELETE - ✓

POST - X (OKAY!)

PUT - ✓ (OVERWRITE)

PATCH - NOT ALWAYS
(GENERALLY,
WE DON'T
USE IT)

ANATOMY OF AN HTTP REQUEST

URL → http://mysite.com/movies/415?year=2015

HTTP METHOD → GET, POST, PUT, PATCH, DELETE, OPTIONS, ETC.

HEADERS → metadata about the req. / key-value pairs

BODY → ACTUAL DATA [JSON, XML, PLAINTEXT, etc.]

→ AUTH, FLAGS, SESSION, CONTENT-TYPE / ACCEPTS,
ORIGIN, USER-AGENT

STATELESS?

- REST APIs maintain NO data between requests!

- CLIENT MUST SEND EVERYTHING w/ EACH REQ

- CERTAIN SEC. PROTOCOLS CAN MAKE THIS DIFF.
TO MAINTAIN

ANATOMY OF A RESPONSE

STATUS CODE → INFO ABOUT THE REQ
SUCCESS / FAILURE

HEADERS

BODY

200 OK

201 CREATED

204 NO CONTENT

400 BAD REQUEST

401 UNAUTHORIZED

403 FORBIDDEN

404 NOT FOUND

500 INTERNAL SERVER ERROR

100-199 INFORMATIONAL

200-299 SUCCESS

300-399 REDIRECT

400-499 CLIENT PROBLEM

500-599 SERVER PROBLEM